

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Semester Winter Examination – Dec. - 2019

Branch: Mechanical Engineering.

Subject: - Applied Thermodynamics – I (BTMECE502)

Date:- 11/12/2019

Sem.:- V

Marks: 60

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

- Q.1 a) What is meant by term fuel? Classify fuels in detail. ----- (6)
b) Explain briefly, the method used to determine the higher calorific value of the liquid. ----- (6)
- Q 2 a) Describe with neat diagram, the construction and working of a Babcock and Wilcox water tube boiler. ----- (6)
b) The following observations were made in a boiler trial:
Coal used 250kg of calorific value 29800kJ/kg, water evaporated 2000kg, steam pressure 11.5 bar, dryness fraction of steam 0.95 and feed water temperature 34°C Calculate the equivalent evaporation "from and at 100°C" per kg of coal and the efficiency of the boiler. ----- (6)
- Q.3 a) With pv and Ts diagram explain Carnot vapour cycle in detail. ----- (6)
b) Show the Rankine cycle on p-v and T-s diagrams and explain the processes involved. ----- (6)
- Q4 a) Dry saturated steam at a pressure of 15 bar enters in a nozzle and is discharged at a pressure – of 1.5 bar. Find the final velocity of steam, when the initial velocity of steam is negligible. If 10% of the heat drop is lost in friction, find the percentage reduction in the final velocity. --- (6)
b) With h-s graph explain effect of friction between nozzle surface and steam. - (6)
- Q.5 a) In a De- laval turbine, the steam enters the wheel through a nozzle with a velocity of 500 m/s and at an angle of 20° to the direction of motion of blade. The blade speed is 200m/s and the exit angle of moving blade is 25°. Find the inlet angle of moving blade, exit velocity of steam, and its direction and work done per kg of steam. ----- (6)
b) What do you mean by compounding of the turbine. Draw the neat sketch of the 3 stage velocity compounding with variation of pressure, velocity and specific volume. (6)
- Q.6 a) With neat sketch and cycle representation explain the working of the centrifugal compressor. (6)
b) With neat sketches explain construction, working of Reciprocating air compressor with applications. (6)

Paper End

Instructions to the Students:

1. All the questions are compulsory.
 2. Use of non-programmable scientific calculators and steam table is allowed.
 3. Assume suitable data wherever necessary and mention it clearly.
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Q. 1 Solve Any Two of the following. (6 x 2) 12 M

- A) Derive the combustion equation for hydrocarbon (solid) fuels
- B) Explain the working of La-Mont Boiler giving its principle and schematic sketch.
- C) Define Calorific value and distinguish between gross calorific value and net calorific value.

Q.2 Solve Any Two of the following. (6 x 2) 12 M

- A) List any two Accessories and Mountings. Explain Economizer in details with the help of neat sketch.
- B) Describe with a neat diagram, the construction and working of a Babcock and Wilcox water tube boiler.
- C) Give the detailed classification of draught. Also explain with neat sketch Induced mechanical draught.

Q. 3 Solve Any Two of the following. (6 x 2) 12 M

- A) Explain ideal Rankine cycle on P-V, T-S & h-s diagram also derive the expression for thermal efficiency of cycle.
- B) Classify steam turbine. Differentiate between impulse and reaction steam turbines
- C) Define steam condenser. Explain surface condenser in detail.

Q.4 Solve Any Two of the following. (6 x 2) 12 M

- A) What is use of cooling tower used in steam power plant? Explain Induced draught cooling tower with neat sketch.
- B) Define nozzle; Also derive expression for condition for maximum discharge through the steam nozzle.
- C) A single acting reciprocating air compressor has cylinder diameter and stroke of 200 mm and 300 mm respectively. The compressor sucks air at 1 bar and 27⁰ C and delivers at 8 bars while running at 100 r.p.m. Find: a. Indicated power of compressor; b. Mass of air delivered by the compressor per minute; c. Temperature of the air delivered by the compressor.

Q. 5 Solve Any Two of the following.

(6 x 2) 12 M

- A) Draw and explain in short P-V diagram of single stage Reciprocating compressor with and without clearance
- B) The following observations were made on a boiler plant during one hour test:
Steam pressure = 20 bar; Steam temperature = 260°C ; Steam generated = 37500 kg;
Temperature of water entering the economizer = 15°C ; Temperature of water leaving the economizer = 90°C ; Fuel used = 4400 kg; Energy of combustion of fuel = 30 000 kJ/kg.
Calculate:
a. The equivalent evaporation per kg;
b. The thermal efficiency of the plant;
c. The percentage heat energy of the fuel energy utilized by the economizer.
- C) With the help of neat sketch explain the working of centrifugal compressor. Also write down the formula to calculate the work done by centrifugal compressor for different processes.

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch : Mechanical Engineering Semester : V Subject Code & Name: Applied Thermodynamics (BTMC506) Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Use of steam table and Mollier diagram is allowed 5. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	How flue gas analysis is done by using Orsat apparatus, Explain by using a neat sketch.	(Understand/ CO1)	6
B)	% composition of liquid fuel by mass is C = 84.8 % , H ₂ = 15.2 % . Calculate i) minimum air required ii) composition of product of combustion if 15 % excess air is supplied.	(Apply /CO1)	6
C)	% volumetric analysis of a sample of flue gases of a coal fired boiler gave CO ₂ = 10.4 % , CO = 0.2 % , O ₂ = 7.8% and N ₂ = 81.6%. Gravimetric analysis of coal was C = 78 % , H ₂ = 6 % , O ₂ = 3 % and incombustible = 13 %. Estimate i) Minimum amount of air required ii) Air fuel ratio	(Apply /CO1)	6
Q.2	Solve Any Two of the following.		12
A)	Distinguish between induced draught and forced draught	(Understand/ CO3)	6
B)	A Lancashire boiler generates 2400 kg of dry saturated steam per hour at a pressure of 11 bar. The grate area is 3 m ² and 100 kg of coal is burnt per m ² of grate area per hour. Calorific value of coal is 32000 kJ/kg and the temperature of feed water is 25 ⁰ C . Determine i) factor of evaporation ii) equivalent evaporation iii) boiler efficiency.	(Apply /CO3)	6
C)	A boiler is equipped with a chimney of 30 meter height. The flue gases which pass through the chimney are at a temperature of 288 ⁰ C, whereas the atmospheric temperature is 18 ⁰ C. If the air flow through the chamber is 18 kg / kg of fuel, find i) theoretical draught in mm of water ii) velocity of flue gases passing through chimney , if 50 % draught is lost in friction.	(Apply /CO3)	6

Q. 3	Solve Any Two of the following.		12
A)	Represent the ideal Reheat cycle on the P-V & T-S diagram and derive the equation for its thermal efficiency.	(Understand/ CO2)	6
B)	A steam power plant is supplied with dry saturated steam at a pressure of 12 bar and exhausts into a condenser at 0.1 bar. Calculate i) turbine work per kg of steam ii) pump work per kg of steam iii) Rankine efficiency.	(Apply /CO2)	6
C)	Dry saturated steam at a pressure of 10 bar enters in a nozzle and with initial velocity of 90 m/s. The outlet pressure is 6 bar and outlet velocity is 435 m/s. The heat loss from the nozzle is 9 kJ/kg of steam flow. calculate the dryness fraction of steam and area at exit if inlet area is 1256 mm ² .	(Apply /CO4)	6
Q.4	Solve Any Two of the following.		12
A)	What are the sources of air leakage in steam condensers ? What is the effect of air leakage on condenser performance and how to eliminate air leakage in condensers ?	(Understand/ CO4)	6
B)	In a surface condenser, the vacuum maintained is 700 mm of Hg. The barometer reads 754 mm of Hg, if the temperature of condensate is 18 °C, determine i) mass of air per kg of steam ii) vacuum efficiency.	(Apply /CO4)	6
C)	Write short note on compounding of steam turbines	(Understand/ CO3)	6
Q. 5	Solve Any two of the following.		12
A)	Derive expression for work done per cycle for the reciprocating air compressor without clearance.	(Understand/ CO5)	6
B)	A single stage , single acting reciprocal compressor has a bore of 200 mm and stroke of 300 mm. It receives air at 1 bar and 20 °C, delivers it at 5.5 bar. If the compression follows law $PV^{1.3}=C$ and clearance is 5 % of stroke , determine , i) mean effective pressure ii) power required to drive the compressor if it runs at 500 rpm.	(Apply /CO5)	6
C)	Differentiate between rotary compressor and reciprocating compressor	(Understand/ CO5)	6
	*** End ***		

Course: B. Tech.

Branch: Mechanical Engineering

Semester: V

Subject Code & Name: BTMEC502 & Applied Thermodynamics I

Max Marks: 60

Date:09/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain the requirements of good fuel	BL ₁	6
B) What is meant by term fuel? Classify fuels in detail	BL ₂	6
C) The volumetric analysis of a gas is given below as, CO ₂ = 14%, CO = 1%, O ₂ = 5%, N ₂ = 80% Obtain the percentage analysis by mass.	BL ₄	6
Q.2 Solve Any Two of the following.		12
A) Describe with neat diagram, the construction and working of a low-pressure boiler with its specifications	BL ₄	6
B) Compare the water tube boiler with fire tube boiler. Also explain the boiler mountings and accessories.	BL ₃	6
C) Explain the Indian Boiler Regulation Act	BL ₃	6
Q. 3 Solve Any Two of the following.		12
A) Explain the Rankine vapour cycle in detail with P-V and T-S graph	BL ₄	6
B) A turbine is supplied with steam at a pressure of 32 bar and a temperature of 410 °C. The steam then expands isentropically to a pressure of 0.08 bar. Find the dryness fraction at the end of expansion Note: Use mollier chart and represent it with neat sketch	BL ₅	6
C) Consider an air standard cycle in which the air enters the compressor at 1.0 bar and 20°C. The pressure of air leaving the compressor is 3.5 bar and the temperature at turbine inlet is 600°C. Determine per kg of air: (i) Efficiency of the cycle, (ii) Heat supplied to air, (iii) Work available at the shaft, (iv) Heat rejected in the cooler, and	BL ₅	6

(v) Temperature of air leaving the turbine.

For air $\gamma = 1.4$ and $c_p = 1.005 \text{ kJ/kg K}$.

Q.4	Solve Any Two of the following.		12
A)	Explain the term compounding of steam turbine. Also draw a sketch of velocity compounding and labelled it.	BL₃	6
B)	A single stage impulse turbine motor has a diameter of 1.2 m running at 3000 rpm. The nozzle angle is 18° , blade speed ratio is 0.42. The ratio of relative velocity at outlet to relative velocity at inlet is 0.9. The outlet blade angle is 30° small than inlet blade angle. The steam flow rate is 5 Kg/sec. Draw a velocity diagram.	BL₅	6
C)	Explain the terms 1. Vacuum Efficiency 2. Condenser efficiency	BL₃	6
Q. 5	Solve Any Two of the following.		12
A)	Classify air compressors. Also state the applications of compressed Air	BL₃	6
B)	Explain the working of single stage reciprocating compressor with indicator (P-V) diagram	BL₄	6
C)	A single acting single stage reciprocating compressor of bore 300 mm and stroke 400 mm is required to compress air from 1 bar to 5 bar. Find the power required to drive the compressor, if it runs at 200 r.p.m. and compression is, Isothermal Adiabatic with index = 1.4 Polytropic with index = 1.3	BL₆	6

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Summer Examination – 2023 Course: B. Tech. Branch: Mechanical Engineering Semester: V Subject Code & Name: BTMC506 & Applied Thermodynamics Max Marks: 60 Date: 22-08-2023 Duration: 3 Hr			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the requirements of good fuel	BL₁	6
B)	What is meant by term fuel? Classify fuels in detail	BL₂	6
C)	The volumetric analysis of a gas is given below as, CO ₂ = 14%, CO = 1%, O ₂ = 5%, N ₂ = 80% Obtain the percentage analysis by mass.	BL₄	6
Q.2	Solve Any Two of the following.		12
A)	Describe with neat diagram, the construction and working of a low-pressure boiler with its specifications	BL₄	6
B)	Compare the water tube boiler with fire tube boiler. Also explain the boiler mountings and accessories.	BL₃	6
C)	Explain the Indian Boiler Regulation Act	BL₃	6
Q. 3	Solve Any Two of the following.		12
A)	Explain the Rankine vapour cycle in detail with P-V and T-S graph	BL₄	6
B)	A turbine is supplied with steam at a pressure of 32 bar and a temperature of 410 °C. The steam then expands isentropically to a pressure of 0.08 bar. Find the dryness fraction at the end of expansion Note: Use mollier chart and represent it with neat sketch	BL₅	6
C)	Consider an air standard cycle in which the air enters the compressor at 1.0 bar and 20°C. The pressure of air leaving the compressor is 3.5 bar and the temperature at turbine inlet is 600°C. Determine per kg of air: (i) Efficiency of the cycle, (ii) Heat supplied to air, (iii) Work available at the shaft, (iv) Heat rejected in the cooler, and	BL₅	6

	(v) Temperature of air leaving the turbine. For air $\gamma = 1.4$ and $c_p = 1.005 \text{ kJ/kg K}$.		
Q.4	Solve Any Two of the following.		12
A)	Explain the term compounding of steam turbine. Also draw a sketch of velocity compounding and labelled it.	BL₃	6
B)	A single stage impulse turbine motor has a diameter of 1.2 m running at 3000 rpm. The nozzle angle is 18° , blade speed ratio is 0.42. The ratio of relative velocity at outlet to relative velocity at inlet is 0.9. The outlet blade angle is 30° small than inlet blade angle. The steam flow rate is 5 Kg/sec. Draw a velocity diagram.	BL₅	6
C)	Explain the terms 1. Vacuum Efficiency 2. Condenser efficiency	BL₃	6
Q. 5	Solve Any Two of the following.		12
A)	Classify air compressors. Also state the applications of compressed Air	BL₃	6
B)	Explain the working of single stage reciprocating compressor with indicator (P-V) diagram	BL₄	6
C)	A single acting single stage reciprocating compressor of bore 300 mm and stroke 400 mm is required to compress air from 1 bar to 5 bar. Find the power required to drive the compressor, if it runs at 200 r.p.m. and compression is, Isothermal Adiabatic with index = 1.4 Polytropic with index = 1.3	BL₆	6
	*** End ***		

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch :Mechanical & Allied Engineering

Semester :V

Subject Code & Name: Applied Thermodynamics (BTMC506)

Max Marks: 60

Date:12/01/2024

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*
- 5. Use of steam table is allowed.*

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain the Orsat apparatus for flue gas analysis with neat schematic diagram.	CO1	6
B) Develop the equation for the measurement of calorific value of solid fuel by the use of bomb calorimeter with neat sketch.	CO1	6
C) The flue gas analysis by volume for burning of C_6H_{14} is as follows: $CO_2 = 8.5\%$, $CO = 7.8\%$, and $N = 83.7\%$ Calculate (a) the air-fuel ratio required for chemically correct combustion.	CO1	6
Q.2 Solve Any Two of the following.		12
A) A boiler generates 8.5 kg of steam per kg of coal burned at a pressure of 13.5 bar from feed water having absolute temperature of 350 K. The boiler efficiency is 70% and factor of evaporation 1.17. Taking specific heat of steam at constant pressure to be 2.1 kJ/kg K, calculate (a) the degree of superheat and temperature of steam generated, (b) the calorific value of coal, and (c) the equivalent evaporation.	CO1	6
B) Define and explain mountings and accessories with suitable any one schematic diagram of both.	CO3	6
C) Derive the expression for the height of natural draught.	CO3	6
Q. 3 Solve Any Two of the following.		12
A) Steam expands from 2.5 bar to 1 bar in a nozzle. The initial velocity of steam is 80 m/s and initial temperature is 200°C. Taking nozzle efficiency as 96%, find the exit velocity.	CO2	6
B) Derive the equation for thermal efficiency of Reheat Rankine cycle with neat sketch.	CO2	6
C) Derive and explain the thermal efficiency of Joule Bryton cycle.	CO2	6
Q.4 Solve Any Two of the following.		12
A) Explain the velocity and pressure compounding of impulse turbine with suitable schematic diagram.	CO4	6

B) In a single stage impulse turbine, the blade angles are equal and the nozzle angle is 20°. The velocity coefficient for the blade is 0.83. Find the maximum blade efficiency possible. If the actual blade efficiency is 90% of maximum blade efficiency, find the possible ratio of blade speed to steam speed.	CO4	6
C) Differentiate the Jet and surface condenser with neat sketch.	CO4	6
Q. 5 Solve Any Two of the following.		12
A) Explain reciprocating air compressor with neat sketch.	CO5	6
B) Explain the benefits of the multi staging in reciprocating air compressor with PV diagram.	CO5	6
C) Derive the equation for indicated work of single stage, single acting reciprocating air compressor with neat sketch.	CO5	6

*** End ***